Civics Group	Index Number	Name (use BLOCK LETTERS)

H1

ST. ANDREW'S JUNIOR COLLEGE
2011 JC2 Preliminary Examinations

H1 BIOLOGY **8875/2**

Paper 2: Core

Friday 16 September 2011 2 hours

Additional Materials: Answer Paper
Cover Sheet for Section B

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and index number on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagram, graph or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A
Answer **all** the questions.

Section B
Answer **one** question.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
Section A	
1	/11
2	/6
3	/11
4	/12
5 or 6	/20
Total	/60

This document consists of **11** printed pages.

[Turn over

Section A

Answer **all** *questions*.

1.

(a) Haemoglobin is an efficient oxygen transport molecule in red blood cells. With reference to its tertiary and quaternary structures, explain how haemoglobin is adapted to its function.

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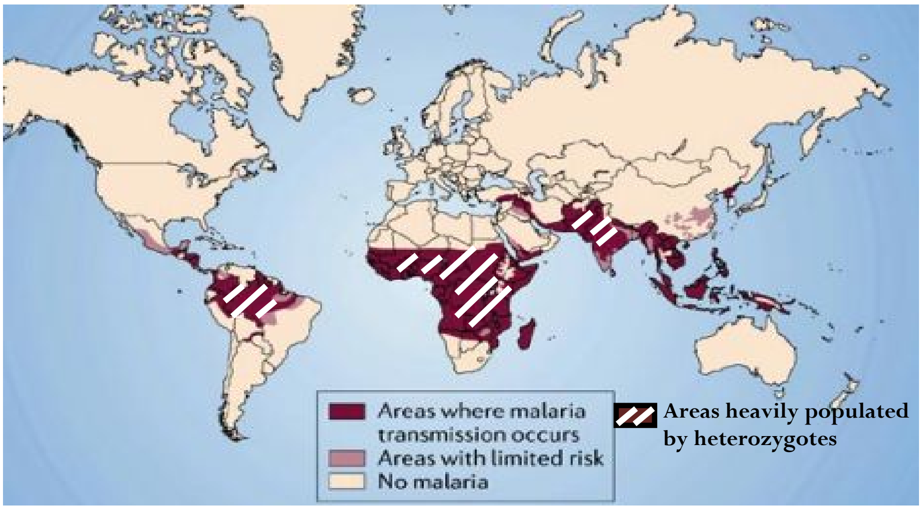
(b) Sickle cell anaemia is a recessive disease whereby the normal haemoglobin (encoded by Hb^A allele) in red blood cells is replaced entirely by abnormal haemoglobin (encoded by Hb^S allele). Abnormal haemoglobin crystallises when oxygen level is low and the crystals causes the red blood cells, which are typically biconcave, to change to a sickle shape. Such sickle shaped red blood cells have a lower lifespan.

Fig. 1.1 shows the areas where malaria is prevalent. Malaria is a disease caused by a *Plasmodium* parasite which spends part of its life cycle inside the red blood cells.



Adapted from http://www.dailymail.co.uk/travel/article-593842/How-avoid-malaria.html

Fig 1.1

(i) Based on Fig. 1.1 and your knowledge on relationship between alleles, suggest explanations for the distribution of heterozygotes.

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Asita and Ayoo is an African couple who are free from symptoms of sickle-cell anemia. Both their parents appear normal but they have siblings who died young from the disease. Asita has blood type O while Ayoo has blood type AB.

Provided that their first child is diseased, use a genetic diagram to calculate the probability that their second child will be a boy with blood type B and normal haemoglobin. State clearly, the types of haemoglobin present in each phenotype. [4]

(ii) Account for the difference in amino acid sequence between normal and abnormal haemoglobin.

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[Qn1 - Total: 11]

2.

(a) Uncontrolled cell division can result in cancer. Some types of cancer can be treated by chemotherapy, which involves the injection of chemicals into the bloodstream.

One chemical used for chemotherapy is called Methotrexate. This drug slows down cell division by acting on one of the enzymes dihydrofolate reductase, in a metabolic pathway.

This enzyme catalyses the conversion of dihydrofolic acid to tetrahydrofolic acid, that results in the formation of purines. The chemical structures of both dihydrofolic acid and Methotrexate are shown below in Fig. 2.1.

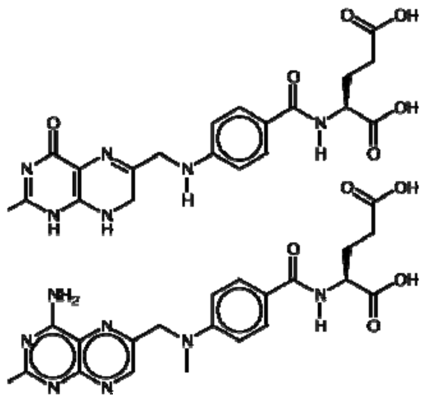

Fig 2.1
Chemical structures of dihydrofolic acid (top) methotrexate (bottom).

Explain how the use of Methotrexate will slow down the mitotic cell cycle.

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(b) Describe briefly how the information on DNA is used to synthesize the RNA transcript for dihydrofolate reductase.

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[Qn2 - Total: 6]

3.
Fig. 3.1 shows the diagram of a mitochondrion and a chloroplast.

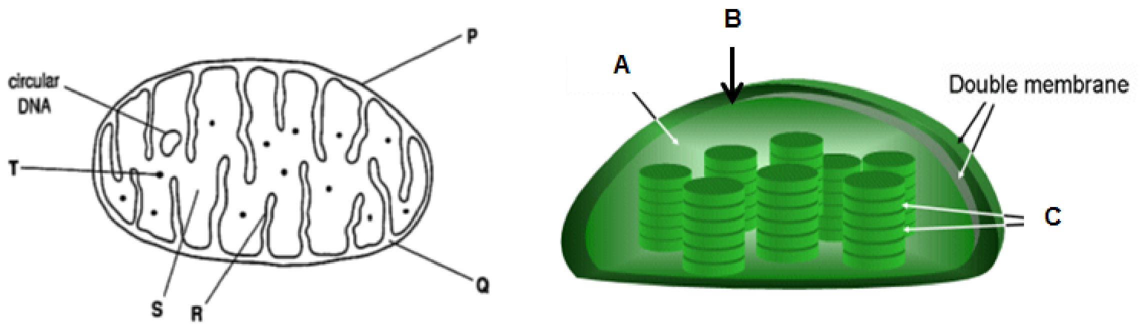

Fig. 3.1

(a) In each case, state the letter(s) that indicate(s) the site of: [2]

(i) Carbon fixation ……………………………………….

(ii) ATP synthesis ………………………………………..

(b) Suggest one function for the circular DNA in mitochondria as shown in Fig. 3.1.

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(c) Briefly outline the process of ATP production in oxidative phosphorylation.

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(d) ATP exists as charged molecules in cells. Suggest why ATP made in the light reaction of photosynthesis can only be used for the dark reaction in the chloroplast and not for other cellular processes.

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(e) Fig.3.2 shows the level of production of the products of photophosphorylation in a plant.

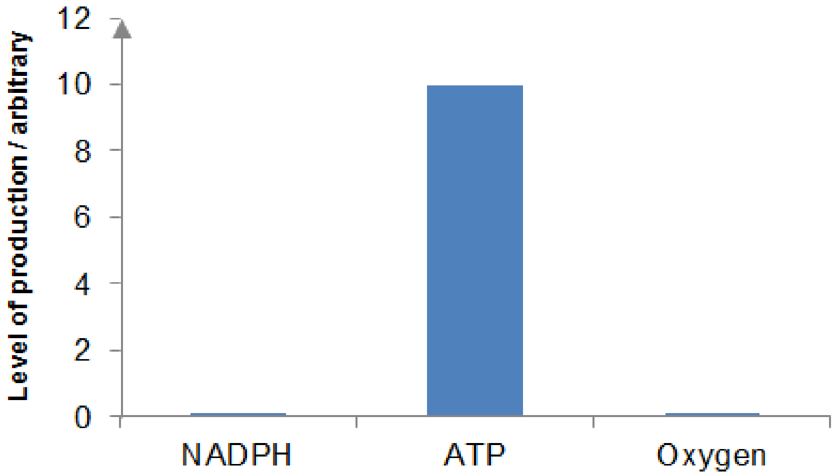

Fig. 3.2

With reference to Fig. 3.2, account for the level of synthesis of NADPH, ATP and oxygen.

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[Qn3 - Total: 11]

4
A molecular scientist is in the process of cloning the human anti-thrombin gene using bacteria cells.

Fig.4.1 shows 2 plasmids (A and B) and the respective restriction sites of various restriction enzymes. In plasmid B, mutations were deliberately introduced and affected restriction sites are denoted by *****. The lengths of the plasmid DNA between consecutive restriction sites are also indicated.

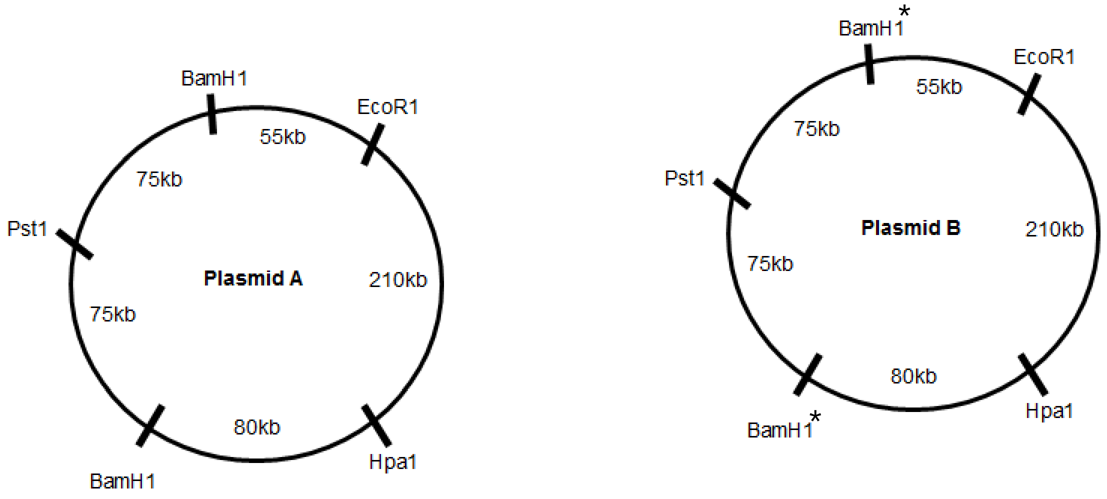

Fig. 4.1

He carried out a series of 2 experiments by adding different restriction enzymes to the plasmids. The components added to each tube can be seen in Table 4.1.

Each tube was then left to incubate at 37ºC for about 15 minutes (sufficient time for complete digestion to take place).

Tube number	Tube Contents
1	Plasmid A + *BamH1* + *Hpa1*
2	Plasmid B + *BamH1* + *EcoR1* + *Pst1*

Table 4.1

(i) Describe the natural role of *BamH1, EcoR1, Pst1* and *Hpa1* in the bacteria in which they are produced.

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(ii) The DNA fragments obtained were then separated by gel electrophoresis. A DNA ladder was loaded into the left most well of the agarose gel. The biggest fragment of the ladder is 550kb and the difference between each fragment of the ladder is 50kb.

The molecular biologist then loaded the mixture from each tube into their respectively numbered wells of the agarose gel as seen in Fig. 4.2.

Predict the results of the gel electrophoresis by drawing in the bands for each of the 2 tubes in Fig. 4.2 below. Indicate the length of each band. [2]

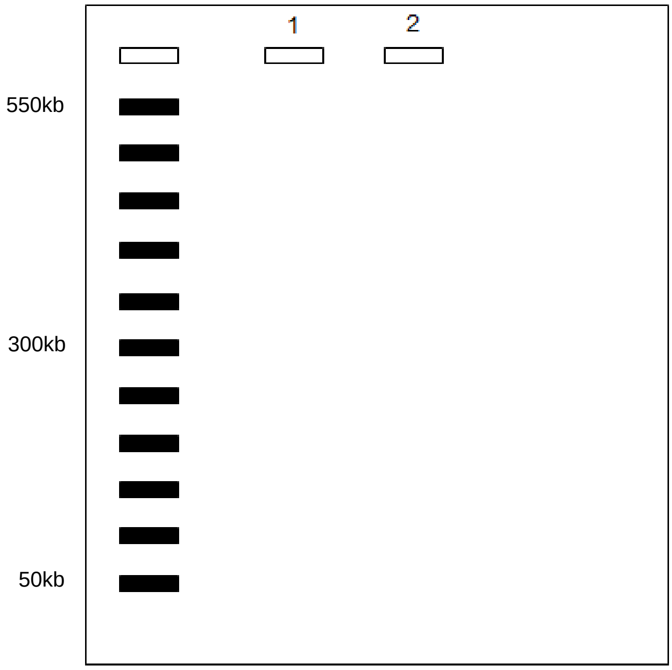

Fig. 4.2.

A genetic engineer cuts plasmid A shown in Fig. 4.1.with *Hpa*I in order to insert a human anti-thrombin gene.

(iii) Outline the steps that must subsequently be followed to produce a recombinant plasmid.

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Fig. 4.3 shows the results of plating the transformed bacteria cells onto an agar plate with the appropriate substances.

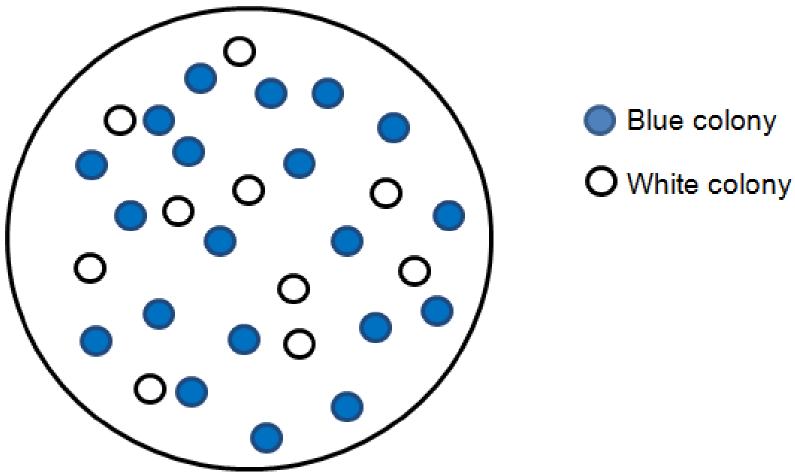

Fig. 4.3

(iv) Calculate the percentage of cells into which the recombinant plasmid had been successfully introduced.

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(v) Explain why some colonies appeared white while others appeared blue.

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(vi) The same molecular scientist has problems obtaining functional anti-thrombin protein in this cloning experiment. Analysis reviewed a much longer amino acid sequence in the non-functional protein. Suggest and explain the reason why.

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[Qn4 - Total: 12]

Section B

Answer **EITHER 5 OR 6** question.

Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections **(a), (b)** etc., as indicated in the question.

Either

5 **(a)** Describe the structure and functions of the centrioles, nucleolus and smooth endoplasmic reticulum in the cell. [8]

(b) Describe the functions of triglycerides and phospholipids in living organisms. [4]

(c) Describe the main features of collagen that contribute to its tensile strength [8]

[Total: 20]

Or

6 **(a)** Describe the development of cancer as a multi-step process. [6]

(b) Describe the unique features of zygotic stem cells, embryonic stem cells and blood stem cells and explain their normal functions in a living organism. [8]

(c) Explain how natural selection may bring about evolution. [6]

[Total: 20]